Supplements

- Supplement 1. Medline OVID search strategy
- Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.
- Supplement 3. Effect sizes for outcome measures and sub-measures
- Supplement 4. Outcome measures utilized in the included studies.

Naehrig D, *et al. BMJ Open* 2021; 11:e046599. doi: 10.1136/bmjopen-2020-046599

Supplement 1. Medline OVID search strategy

1. general practitioners/
2. "general practitioner*".mp.
3. (GP or GPs).mp.
4. exp General Practice/
5. Family Practice/ or Primary Health Care/
6. "family practitioner*".mp.
7. "primary care practitioner*".mp.
8. physicians, family/ or physicians, primary care/
9. "family medicine".mp.
10. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9
11. ("psychological well being" or "psychological wellbeing").mp.
12. (wellbeing or "well being").mp.
13. ("cognitive well being" or "cognitive wellbeing").mp.
14. flourishing.mp.
15. ((job or work) adj3 (satisf* or engage* or motivat*)).mp.
16. life satisfaction*.mp.
17. Job Satisfaction/
18. resilien*.mp.
19. Resilience, Psychological/
20. burnout.mp.
21. Burnout, Professional/
22. Compassion fatigue/
23. compassion fatigue.mp.
24. joy.mp.
25. "joy in practice".mp.
26. 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25
27. Motivational interviewing/
28. Mindfulness/
29. Mediation/

30. mindful*.mp.

31. meditat*.mp.

32. Self-Care/

33. ("self care" or "self-care").mp.

34. Motivation/

35. Professional autonomy/

36. ("self determination" or "self-determination").mp.

37. exp Randomised Controlled Trial/ or Clinical Trial/

38. trial.mp.

39. Early Medical Intervention/ or intervention.mp.

40. Counseling/

41. Course?ling.mp.

42. Mentoring/

43. mentor*.mp.

44. Education/

45. education.mp.

46. workshop*.mp.

47. training.mp.

48. coaching.mp.

49. facilitat*.mp.

50. 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49

51. 10 and 26 and 50

Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.

Within-group:

Mean_change_WG = Mean_Exp_post – Mean_Exp_pre

SD* pooled _WG = SQRT(((N_Exp_post-1) * SD_Exp_post^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Exp_post + N_Exp_pre – 2))

ES_WG = Mean_change_WG / SD* pooled_WG

Between-group:

Mean_change_BG = (Mean_Exp_post – Mean_Exp_pre) – (Mean_Ctrl_post – Mean_Ctrl_pre)

SD* pooled_BG = SQRT(((N_Ctrl_pre-1) * SD_Ctrl_pre^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Ctrl_pre + N_Exp_pre – 2))

ES_BG = Mean change_BG / SD* pooled_BG

Legend. ES = Effect size, WG = Within group, BG = Between group, Exp = experimental group, Ctrl = control group, N = sample size, pre = pre-intervention measure, post = post-intervention measure, SQRT = square root, ^2 = raised to the power of two.

Supplement 3. Effect sizes for outcome measures and sub-measures

Name, n pre/post	Measure	Within group absolute mean change (post - pre experiment)	SD* Exp pooled	ES Hedges' g Exp (Delta mean/ SD* pool)	Between group absolute mean change (experiment – control group)	SD* Baseline pooled	ES Hedges' g (Delta Mean / SD* pool)
With control group							
AMUTIO*	FFMQ	0.37	0.48	0.78	0.54	0.36	1.51
n=21/21	Observe	0.65	0.62	1.05	0.84	0.75	1.12
	Describe	0.25	0.67	0.37	0.31	0.63	0.49
	Awareness	0.32	0.77	0.42	0.34	0.79	0.43
	Non-judgement	0.36	0.65	0.55	0.39	0.58	0.67
	Non-reactive	0.29	0.57	0.51	0.26	0.54	0.48
	SRSI-3 Relaxation	0.54	0.57	0.95	0.46	0.52	0.89
	SRSI-3 Pos energy	0.71	0.74	0.97	0.71	0.63	1.13
	SRSI-3 Mindfulness	0.71	0.81	0.88	0.76	0.91	0.84
	SRSI-3 Transcend	0.6	0.93	0.65	0.84	0.88	0.96
VERWEIJ**	FFMQ	6.87	14.49	0.47	6.9	13.82	0.50
n=43/43 exp,							
n=20/20 control	Observe	2.04	4.49	0.45	1.56	4.28	0.36
	Describe	1.51	4.69	0.32	1.64	4.52	0.36
	Awareness	0.98	4.59	0.21	1.57	4.47	0.35
	Non-judgement	1.2	5.25	0.23	1.37	5.04	0.27
	Non-reactive	1.13	3.71	0.30	0.76	3.56	0.21
	JSPE total	1.95	9.90	0.20	0.2	9.61	0.02
ASSUERO	FFMQ	12.40	19.50	0.64	11.3	19.65	0.57
n=43/43 exp							
n=25/25 control	Observe	4.30	7.07	0.61	4.7	7.33	0.64
	Describe	0.70	5.50	0.13	0.6	5.52	0.11
	Awareness	2.10	5.33	0.39	2.7	6.24	0.43
	Non-judgement	3.50	7.05	0.50	2.4	7.35	0.33
	Non-reactive	2.20	3.71	0.59	1.8	3.85	0.47
	JSPE total	3.50	11.32	0.31	5.3	12.10	0.44
	Perspective	1.30	5.88	0.22	1.5	6.62	0.23
	Compassion	1.80	4.54	0.40	2.3	5.03	0.46
	In patients shoes	0.40	1.49	0.27	1.5	1.54	0.98
DYRBYE	CD-RISC-10	1.30	5.44	0.24	0.7	5.46	0.13
n=40/40 exp							
n= 39/39 control	PJSS	1.00	9.94	0.10	0.6	10.11	0.06
	UWES vigor	0.30	0.94	0.32	0.2	1.04	0.19
	UWES dedication	0.10	0.87	0.11	-0.1	0.96	-0.10
	UWES absorption	-0.10	0.88	-0.11	-0.2	1.00	-0.20
	Empowerment at work	2.7	11.29	0.24	-0.3	12.56	-0.02
GARDINER	WRM-7 total	2.79	6.46	0.43	2	6.63	0.30
n=86/77 exp,							
n=24/19 control	QoWL-6 total	2.92	6.50	0.45	1.76	6.48	0.27
SCHROEDER	MAAS	0.62	0.99	0.63	0.76	0.86	0.88
N=16/13 exp,							
n=17/13 control	BRS	2.53	4.93	0.51	2.95	4.81	0.61
	SCBC	1.53	4.33	0.35	3.46	4.75	0.73
MARGALIT***							
n=44/44	Self-esteem	4.4	13.72	0.32			

Without control group					
MONTERO-MARIN					
n=30/30	PANAS 1 session/wk	1.25	6.10	0.20	
	PANAS 2 session/wk	2.97	5.69	0.52	
	MAAS 1 session/wk	2.67	11.49	0.23	
	MAAS 2 session/wk	4.59	12.28	0.37	
	CD-RISC total 1 session/wk	1.23	7.31	0.17	
	CD-RISC total 2 session/wk	2.48	6.79	0.37	
VON WIETMARSCHEN	SCS	0.5	0.65	0.77	-
n=50/50	GRAS	3.3	7.22	0.46	-
FORTNEY	RS-14	2.1	12.28	0.17	-
n=30/28	SCBC	-0.2	4.54	-0.04	-
KRASNER	JSPE total	4	9.16	0.44	-
n=60/59	Perspective	2	5.79	0.35	-
	Compassion	1.6	4.24	0.38	-
	In patients' shoes	0.8	2.12	0.38	-
	Mindfulness 2 factor FFMQ (BAER) total	7.7	7.33	1.05	-
	BAER observe	5	4.63	1.08	-
	BAER non-react	3.2	3.95	0.81	-
REES	WHO-5 (WB)	10.3	19.79	0.52	-
n=7/7	PANAS-pos	-			-
WHALLEY n=2081/1345	JSS WCW	0.59	1.35	0.44	-

Notes: Amutio*, no information given on pre-post sample size, assumption made that there was no attrition. Verweij**, wait list control participants were included in both intervention and control groups. *** Margalit only pooled group (experimental and control) analysis provided. For the FFMQ Verweij and Assuero have much larger numbers than Amutio, this is presumably due to reporting of the total score count as opposed to the total mean scores.

Outcome measures: JSS WCW Warr Cook Wall Job satisfaction Scale; JSPE Jefferson Scale of Physician Empathy; FFMQ Five Facet Mindfulness Questionnaire; Baer (2 item version of FFMQ); SRSI Smith Relaxation States Inventory; SCS Self Compassion Scale (Neff); FR-JS Font Roja Job Satisfaction Questionnaire; SCBC Santa Clara brief compassion scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; BRS Brief Resilience Scale; GRAS Groningen Reflection Ability Scale; RS-14 Resilience, WHO-5 Wellbeing Index, WRM Work Related Morale Measure. QoWL-6 Quality of Work Life.

Supplement 4. Outcome measures utilized in the included studies.

Acronym	Outcome measure	No of studies used in
ACP/ASIM	Physician satisfaction (Am College of Phys / Am Soc of Int Med)	1
BAER	Baer mindfulness scale (2 item version of FFMQ)	1
BRS	Brief resilience scale	1
CD-RSIC-10	Connor-Davidson Resilience Scale	2
FFMQ	Five Facet Mindfulness Questionnaire	3
FR-JS	Font Roja Job Satisfaction scale	1
GRAS	Groningen Reflection Ability Scale	1
JSPE	Jefferson Scale of Physician Empathy	3
JSS WCW	Job Satisfaction Scale Warr-Cook-Wall	2
LS	Life satisfaction	1
MAAS-15	Mindful attention awareness scale	2
PANAS	Positive and negative affect scale	2
PJSS	Physician Job satisfaction scale	1
PWS (MEMO)	Physician Workplace Satisfaction	2
QoWL-6	Quality of work life	1
RS-14	Resilience	1
SCBC	Santa Clara brief compassion scale	2
SCS	Self compassion scale (Neff)	1
SRSI-3	Smith Relaxation states	1
UWES	Utrecht work engagement score	1
WHO-5	Wellbeing Index	1
WRM-7	Work related morale measure	1
	Self-esteem	1